

**EXAMPLE COMPUTING SYSTEM, SOFTWARE ARCHITECTURE, AND IMPLEMENTATION DETAILS**

**A. Overview**

In various embodiments, the disclosed system includes one or more computing components configured to receive node data, perform signal processing, generate privacy-preserving metrics, perform cross-device correlation and cross-user correlation, and present outputs via a user interface. Processing may be performed on-node, on a user device, on a remote server, or in a distributed configuration.

**B. Example Hardware Components**

In certain embodiments, each node includes: 1. a power subsystem comprising a battery and charging circuitry; 2. one or more sensors and associated analog front ends; 3. a controller or microcontroller configured to sample sensors and manage power; 4. memory configured to store samples, timestamps, and/or derived features; and 5. a communications subsystem configured for wireless communication. In certain embodiments, a user device comprises a mobile phone or tablet including a processor, memory, a display, and wireless connectivity. In certain embodiments, a server environment comprises one or more processors, memory, storage, and networking configured to perform computation and store data.

**C. Example Software Modules**

In various embodiments, the computing system includes software modules comprising: 1. a device manager configured to pair, authenticate, and manage node communications; 2. a time synchronization module configured to align streams from multiple nodes; 3. a signal conditioning module configured to filter, normalize, and temperature-compensate raw signals; 4. an artifact rejection module configured to compute signal quality indices and reject invalid segments; 5. a feature extraction module configured to compute features per sensor modality and time window; 6. a metric generation module configured to generate privacy-preserving indices, zones, ranges, and curves; 7. a cross-device correlation module configured to compute correlation scores, lag signatures, and context-tagged correlation; 8. a cross-user correlation module configured to compute privacy-preserving alignment, synchrony, and stability metrics under consent gating; and 9. a user interface module configured to present outputs and confidence indicators without explicit anatomical imagery.

**D. Data Structures and Storage**

In certain embodiments, the system stores data as time-stamped records including raw samples, conditioned signals, features, and/or metrics. In some embodiments, raw samples are retained only temporarily and then discarded after feature extraction. In various embodiments, the system stores derived metrics and quality indicators to support longitudinal tracking and cross-session comparison.

**E. Model and Rule Execution**

In certain embodiments, the system uses rule-based processing to compute quality indices, detect stable windows, and generate preliminary metrics. In other embodiments, the system uses statistical models and/or machine learning models to generate features, embeddings, and correlation metrics. Models may be individualized using baseline acquisition results and user history. In some embodiments, model outputs include confidence measures and may be constrained to privacy-preserving representations.

## **F. User Interface and Presentation**

In various embodiments, the user interface presents outputs as abstract indices, ranges, zones, curves, trend views, and confidence indicators. The interface may provide time-window navigation, within-user comparisons, and cross-device correlation views. In certain embodiments, the interface supports cross-user correlation views with consent indicators, confidence indicators, and explanatory tags describing whether results are based on matched context windows.

## **G. Update, Calibration, and Maintenance**

In certain embodiments, nodes support firmware updates and calibration routines. Calibration may include sensor offset calibration, temperature compensation calibration, optical coupling checks, electrode impedance checks, and/or geometry reference checks. The computing system may support versioning of processing pipelines and models to ensure consistency of longitudinal tracking.

## **H. Variations**

The above computing architecture is non-limiting. Modules may be combined, separated, reordered, or distributed differently. Certain embodiments operate entirely offline on a user device; other embodiments use cloud computation with access controls. Certain embodiments compute only privacy-preserving metrics and do not store raw signals.